Assignment 2

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Part (a): 4×3 Grid World Reachability and Probabilities

Starting at state $(1, 1)$ with $P((1, 1)) = 1.0$, executing the action sequence [Up, Up, Right, Right, Right] yields the following step-by-step probability distribution.

Step-by-Step State Distributions

Step 1: Action = Up

- Up $\rightarrow (1, 2)$ with probability 0.8
- Left $\rightarrow$ hits wall $\rightarrow (1, 1)$ with probability 0.1
- Right $\rightarrow (2, 1)$ with probability 0.1

Step 2: Action = Up

- From $(1, 2)$: Up $\rightarrow (1, 3)$ (0.64), Left/Right into wall/obstacle $\rightarrow (1, 2)$ (0.16)
- From $(1, 1)$: Up $\rightarrow (1, 2)$ (0.08), Left $\rightarrow (1, 1)$ (0.01), Right $\rightarrow (2, 1)$ (0.01)
- From $(2, 1)$: Up into obstacle $\rightarrow (2, 1)$ (0.08), Left $\rightarrow (1, 1)$ (0.01), Right $\rightarrow (3, 1)$ (0.01)

State distribution: $(1, 3) : 0.64$, $(1, 2) : 0.24$, $(2, 1) : 0.09$, $(1, 1) : 0.02$, $(3, 1) : 0.01$

Step 3: Action = Right

State distribution: $(2, 3) : 0.512$, $(1, 2) : 0.258$, $(1, 3) : 0.088$, $(3, 1) : 0.073$, $(2, 1) : 0.034$, $(1, 1) : 0.026$, $(4, 1) : 0.008$, $(3, 2) : 0.001$

Step 4: Action = Right

State distribution: $(3, 3) : 0.4097$, $(1, 2) : 0.2178$, $(2, 3) : 0.1728$, $(4, 1) : 0.0656$, $(1, 3) : 0.0346$, $(3, 1) : 0.0346$, $(1, 1) : 0.0284$, $(2, 1) : 0.0276$, $(3, 2) : 0.0073$, $(4, 2) : 0.0016$

Step 5: Action = Right

Grid Cell	Exact Fraction	Decimal Probability	Percentage
$(4,3)$	$\frac{4099}{12500}$	0.32792	32.792% (Goal)
$(1,2)$	$\frac{9027}{50000}$	0.18054	18.054%
$(3,3)$	$\frac{8997}{50000}$	0.17994	17.994%
$(4,1)$	$\frac{543}{6250}$	0.08688	8.688%
$(2,3)$	$\frac{389}{6250}$	0.06224	6.224%
$(3,2)$	$\frac{4443}{100000}$	0.04443	4.443%
$(2,1)$	$\frac{353}{12500}$	0.02824	2.824%
$(3,1)$	$\frac{2627}{100000}$	0.02627	2.627%
$(1,3)$	$\frac{631}{25000}$	0.02524	2.524%
$(1,1)$	$\frac{1231}{50000}$	0.02462	2.462%
$(4,2)$	$\frac{171}{12500}$	0.01368	1.368% (Penalty)

Every non-obstacle cell in the grid can be reached, with the intended goal state $(4,3)$ having the highest probability (32.792%).

Part (b): MDP Transformations & Linear Property

i. Transforming $R(s,a,s')$ into $R(s,a)$

Define the expected immediate reward function $R'(s,a)$ as:

$$R'(s,a) = \sum_{s'} P(s' \mid s,a)\, R(s,a,s')$$

Proof of Correspondence. The Bellman expectation equation for any fixed policy π under $R(s,a,s')$ is:

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s) \sum_{s'} P(s' \mid s,a) \left[R(s,a,s') + \gamma V^{\pi}(s')\right]$$

Distributing the sum:

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s) \left[\underbrace{\sum_{s'} P(s' \mid s,a) R(s,a,s')}_{R'(s,a)} + \gamma \sum_{s'} P(s' \mid s,a) V^{\pi}(s') \right]$$

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s) \left[R'(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V^{\pi}(s') \right]$$

Because $V^{\pi}(s)$ is identical under both formulations for every policy π, the expected return for all policies is preserved. Thus, $\pi^* = \arg\max_{\pi} V^{\pi}(s)$ is identical in both MDPs.

ii. Transforming $R(s, a)$ into $R(s)$

To remove action dependence from the reward function, augment the state space by introducing intermediate state-action states:

- Define a new state space $S' = S \cup (S \times A)$.
- When the agent in state $s \in S$ chooses action a, it transitions deterministically to the intermediate state $s_a = (s, a)$ with reward $R(s) = 0$.
- From state $s_a = (s, a)$, the environment automatically transitions to state $s' \in S$ according to $P(s' \mid s, a)$ with state reward $R(s_a) = R(s, a)$.

Under this construction, every policy in the original MDP maps to an equivalent policy in the transformed MDP. The cumulative discounted rewards match up to a constant discount factor scaling, preserving the exact ordering and optimal policies π^*.

iii. Proof that $V^\pi(s)$ Varies Linearly with $R(s)$

Let R be the column vector of state rewards $R(s)$, V^π be the value vector for policy π, and P^π be the transition probability matrix under π. The Bellman equation in vector form is:

$$V^\pi = R + \gamma P^\pi V^\pi \quad \implies \quad (I - \gamma P^\pi)V^\pi = R$$

Since $\gamma < 1$ and P^π is a row-stochastic matrix, $(I - \gamma P^\pi)$ is invertible:

$$V^\pi = (I - \gamma P^\pi)^{-1} R$$

Let $M = (I - \gamma P^\pi)^{-1}$. For any individual state s:

$$V^\pi(s) = \sum_{s'} M_{s,s'} R(s')$$

This is a linear combination of the components of R. Thus, $V^\pi(s)$ varies linearly with $R(s)$.

Part (c): 3-State Undiscounted MDP

i. Qualitative Analysis of Optimal Policy

State 1 ($R(1) = -1$): Action b provides a direct path to the terminal state 3 with probability 0.1. Action a moves the agent to state 2, which incurs a harsher penalty per step ($R(2) = -2$). Thus, state 1 should select action b to exit as quickly as possible.

State 2 ($R(2) = -2$): Taking action b gives an expected value of $V(2) = -2 + 0.9V(2) \implies V(2) = -20$. Taking action a moves the agent with probability 0.8 to state 1, where cost per step is lower (-1) and escape is faster. Thus, state 2 should select action a to transfer to state 1.

ii. Policy Iteration starting with $\pi_0 = (b, b)$

Iteration 1

Policy Evaluation of $\pi_0 = (b, b)$:

$$V^{\pi_0}(1) = -1 + 0.9V^{\pi_0}(1) + 0.1(0) \implies 0.1V^{\pi_0}(1) = -1 \implies \mathbf{V^{\pi_0}(1) = -10}$$
$$V^{\pi_0}(2) = -2 + 0.9V^{\pi_0}(2) + 0.1(0) \implies 0.1V^{\pi_0}(2) = -2 \implies \mathbf{V^{\pi_0}(2) = -20}$$

Policy Improvement:

$$Q(s, a) = R(s) + \sum_{s'} P(s' \mid s, a)\, V^{\pi_0}(s')$$

State 1:

$$Q(1, a) = -1 + 0.2(-10) + 0.8(-20) = -19$$
$$Q(1, b) = -1 + 0.9(-10) + 0.1(0) = -10$$
$$\pi_1(1) = \arg\max\{-19, -10\} = b$$

State 2:

$$Q(2, a) = -2 + 0.8(-10) + 0.2(-20) = -14$$
$$Q(2, b) = -2 + 0.9(-20) + 0.1(0) = -20$$
$$\pi_1(2) = \arg\max\{-14, -20\} = a$$

New policy: $\pi_1 = (b, a)$.

Iteration 2

Policy Evaluation of $\pi_1 = (b, a)$:

$V^{\pi_1}(1) = -1 + 0.9V^{\pi_1}(1) \implies \mathbf{V^{\pi_1}(1) = -10}$

$V^{\pi_1}(2) = -2 + 0.8V^{\pi_1}(1) + 0.2V^{\pi_1}(2) \implies 0.8V^{\pi_1}(2) = -2 + 0.8(-10) = -10 \implies \mathbf{V^{\pi_1}(2) = -12.5}$

Policy Improvement:

State 1:

$$Q(1, a) = -1 + 0.2(-10) + 0.8(-12.5) = -13$$
$$Q(1, b) = -1 + 0.9(-10) = -10$$
$$\pi_2(1) = \arg\max\{-13, -10\} = b$$

State 2:

$$Q(2, a) = -2 + 0.8(-10) + 0.2(-12.5) = -12.5$$
$$Q(2, b) = -2 + 0.9(-12.5) = -13.25$$
$$\pi_2(2) = \arg\max\{-12.5, -13.25\} = a$$

Since $\pi_2 = \pi_1 = (b, a)$, Policy Iteration converges.

Optimal Policy: $\pi^*(1) = b,\ \pi^*(2) = a$
Optimal Values: $V^*(1) = -10,\ V^*(2) = -12.5$

iii. Initial Policy $\pi_0 = (a, a)$, Discounting, and Dependency on γ

Initial Policy $\pi_0 = (a, a)$. In the undiscounted case ($\gamma = 1$), the policy (a, a) cycles indefinitely between state 1 and state 2 without any probability of reaching state 3. The matrix $(I - P)$ is singular, making $V(1)$ and $V(2)$ equal to $-\infty$ (unbounded cost). Standard policy evaluation fails.

Does Discounting Help? Yes. For $\gamma < 1$, geometric discounting guarantees convergence even for infinite trajectories:

$$V(1) = -1 + \gamma\big(0.2V(1) + 0.8V(2)\big)$$
$$V(2) = -2 + \gamma\big(0.8V(1) + 0.2V(2)\big)$$

This yields finite values for $V(1)$ and $V(2)$, enabling the policy improvement step to properly calculate $Q(s, b)$ and update the policy to (b, a).

Does the Optimal Policy Depend on γ? No. For any $\gamma \in (0, 1]$:

$$V^*(1) = \frac{-1}{1 - 0.9\gamma}$$
$$V^*(2) = \frac{-2 + 0.6\gamma}{1 - 0.2\gamma - 0.72\gamma^2 + 0.18\gamma^2}$$

Evaluating

$$Q(1, b) - Q(1, a) = \frac{0.8\gamma(1 - 0.8\gamma)}{(1 - 0.2\gamma)(1 - 0.9\gamma)} > 0$$

Evaluating

$$Q(2, a) - Q(2, b) = \frac{0.6\gamma}{1 - 0.2\gamma} > 0$$

The optimal policy remains $\pi^* = (b, a)$ for all valid discount factors $\gamma \in (0, 1]$.

Part (d): Parametric Value Function Approximation & TD Updates

Given the value function approximator:

$$\hat{V}_\theta(x, y) = \theta_0 + \theta_1 x + \theta_2 y + \theta_3 \sqrt{(x - x_g)^2 + (y - y_g)^2}$$

The Temporal Difference (TD) error δ is defined as:

$$\delta = R(x, y) + \gamma \hat{V}_\theta(x', y') - \hat{V}_\theta(x, y)$$

Using the Widrow–Hoff gradient descent update rule:

$$\theta_i \leftarrow \theta_i + \alpha\, \delta\, \frac{\partial \hat{V}_\theta(x, y)}{\partial \theta_i}$$

Partial Derivatives and TD Update Equations

For θ_0:

$$\frac{\partial \hat{V}_\theta(x,y)}{\partial \theta_0} = 1 \implies \boldsymbol{\theta_0 \leftarrow \theta_0 + \alpha\delta}$$

For θ_1:

$$\frac{\partial \hat{V}_\theta(x,y)}{\partial \theta_1} = x \implies \boldsymbol{\theta_1 \leftarrow \theta_1 + \alpha\delta x}$$

For θ_2:

$$\frac{\partial \hat{V}_\theta(x,y)}{\partial \theta_2} = y \implies \boldsymbol{\theta_2 \leftarrow \theta_2 + \alpha\delta y}$$

For θ_3:

$$\frac{\partial \hat{V}_\theta(x,y)}{\partial \theta_3} = \sqrt{(x-x_g)^2 + (y-y_g)^2} \implies \boldsymbol{\theta_3 \leftarrow \theta_3 + \alpha\delta\sqrt{(x-x_g)^2 + (y-y_g)^2}}$$